

*MA.8.GR.2.2***Benchmark**

MA.8.GR.2.2 Given a pre-image and image generated by a single dilation, identify the scale factor that describes the relationship.

Benchmark Clarifications:

Clarification 1: Instruction includes the connection to scale drawings and proportions.

Clarification 2: Instruction focuses on the preservation of similarity and the lack of preservation of congruence when a figure maps onto a scaled copy of itself, unless the scaling factor is 1.

Connecting Benchmarks/Horizontal Alignment

- MA.8.GR.2.1, MA.8.GR.2.3

Terms from the K-12 Glossary

- Dilation
- Scale Factor

Vertical Alignment**Previous Benchmarks**

- MA.7.GR.1.5

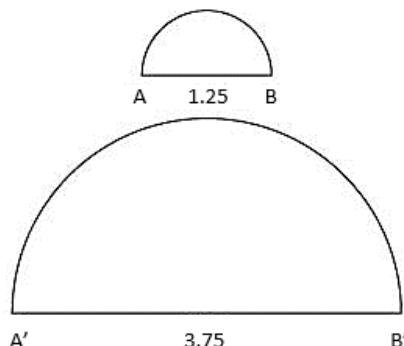
Next Benchmarks

- MA.912.F.2.1, MA.912.F.2.2
- MA.912.GR.2.1, MA.912.GR.2.3, MA.912.GR.2.4

Purpose and Instructional Strategies

In grade 7, students solved mathematical and real-world problems involving scale factors. In grade 8, students determine the scale factor that describes the relationship after a single dilation. In Geometry, students will use dilations to study similarity.

- Instruction includes the use of real-world examples that do not have to be a geometric figure.
 - For example, projections, photocopies and maps have images generated by a single dilation.
- Students will need to understand scale factor to help in the idea of an image enlarging or reducing. A scale factor between 0 and 1 will be a reduction in the image. A scale factor that is greater than one will result in an enlargement of the image.
- Transformations can be noted using the prime notation (') for the image and its vertices. The preimage and its vertices will not have prime notation.
 - For example, the picture below showcases a single dilation.



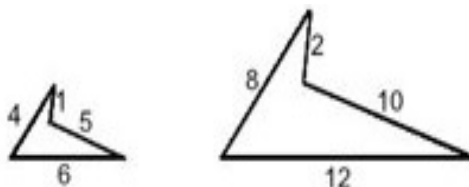
- The expectation of this benchmark is not to represent a dilation on the coordinate plane as this will be included in MA.8.GR.2.3 instruction.

Common Misconceptions or Errors

- Students may incorrectly visualize the images with scale factors. To address this misconception, include practice in visualizing the reduction and enlargement based on the scale factors.

Strategies to Support Tiered Instruction

- Instruction includes practice in visualizing the reduction and enlargement based on the scale factors for students that incorrectly visualize images with scale factors.
 - For example, two figures that have the same shape are said to be similar. When two figures are similar, the ratios of the lengths of their corresponding sides are equal and corresponding angles are congruent. Similar figures have the same shape, but not necessarily the same size.



- Teachers can help students understand that two figures that have the same shape are said to be similar. When two figures are similar, the ratios of the lengths of their corresponding sides are equal. In example 1, the corresponding sides are 8:4, 2:1, 10:5, and 12:6. These ratios are equal to 2, meaning the shapes must be similar. The figures do not have to be the same size to be similar.
- Instruction includes practice in visualizing the reduction and enlargement based on the scale factors.

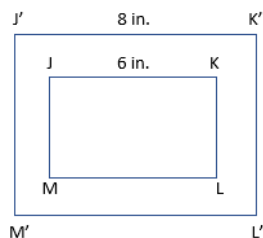
Instructional Tasks

Instructional Task 1 (MTR. 7.1)

The height of a document on your computer is 20 centimeters. When you change the setting to zoom in or out, you change it from 100% to 25%. The new image of your document is a dilation of your original document, the preimage. Determine the scale factor and the height of the new image.

Instructional Items*Instructional Item 1*

Does the image show reduction or enlargement from the quadrilateral $JKLM$? What is the scale factor?



**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*